

Circuit analysis

Probability

We will need three probability rules:

- **Complementary rule:** $P(A) = 1 - P(\text{not } A)$, eg

$$P(\text{item works}) = 1 - P(\text{item doesn't work})$$

- **General addition rule (GAR):** $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

- **Basic multiplication rule (BMR):**

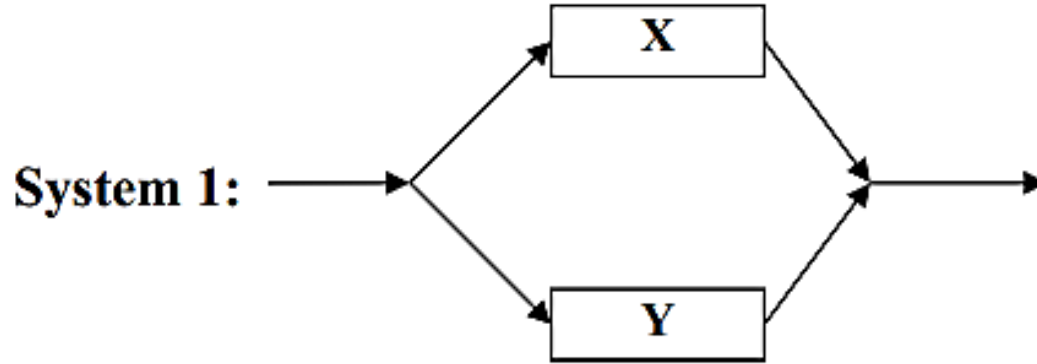
$$P(A \text{ and } B) = P(A) P(B), \text{ when } A, B \text{ are independent}$$

when to use/write **AND** and when to write **OR** is key, so to make this extra clear: You should be able to put in the word **BOTH** with the **AND**, e.g

BOTH A AND B; If this makes sense then use the basic multiplication rule.

You should be able to put the word **EITHER** with **OR**, e.g

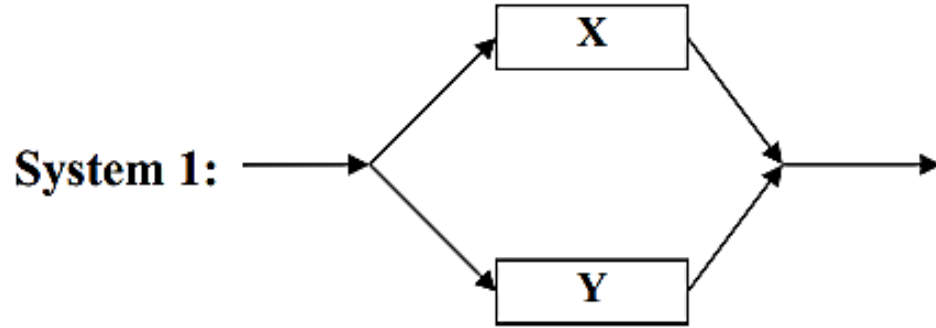
EITHER A OR B.



Imagine yourself on the left hand side of system 1. To get to the other side you have to follow the path and go through EITHER X OR Y, so this will be a general addition rule, since we have “EITHER ... OR ...”.



Imagine yourself on the left hand side of system 2. To get to the other side you have to follow the path and go through **BOTH A AND B**, so this will be a basic multiplication rule, since we have “BOTH ... AND ...”



Say the probability that X works is 0.8, and the probability that Y fails is 0.1, and we want to calculate the probability that the system works. X, Y are independent

When you end up with an “OR” in your statement of the system then you will use the “general addition rule

So,

$$P(\text{System works}) = P(\text{(EITHER) X works OR Y works})$$

and using the general addition rule this becomes:

$$= P(X \text{ works}) + P(Y \text{ works}) - P(X \text{ works AND Y works}) \quad \text{[General addition rule - GAR]}$$

We have the probability that Y fails as 0.1, so we use the complementary rule to tell us

$$P(Y \text{ works}) = 1 - P(Y \text{ doesn't work}) \text{ which gives } P(Y \text{ works}) = 1 - 0.1 = 0.9.$$

Always watch to see if the question is telling you the probability that you want, or if you need to apply the complementary rule first. Anyway, back to the problem ...

$$= 0.8 + 0.9 - P(X \text{ works AND Y works})$$

So long as “X works”, “Y works” are independent then we can apply the basic multiplication rule to the last part of this. You will find in the wording for all of the questions we will deal with that this will be the case, so we can use the basic multiplication rule to get:

$$= 0.8 + 0.9 - P(X \text{ works}) * P(Y \text{ works})$$

$$= 0.8 + 0.9 - 0.8 * 0.9$$

$$= 0.98.$$



Say the probability that A fails is 0.15, and the probability that B fails is 0.18, and we want to calculate the probability that the system works. A, B are independent

$$P(\text{System works}) = P(\text{(BOTH) A works } \underline{\text{AND}} \text{ B works})$$

$$P(\text{A works}) = 1 - P(\text{A fails}) = 1 - 0.15 = 0.85$$

$$P(\text{B works}) = 1 - 0.18 = 0.82$$

Fill these in, to get:

$$\begin{aligned} P(\text{System works}) &= 0.85 * 0.82 \\ &= 0.697 \end{aligned}$$

In summary:

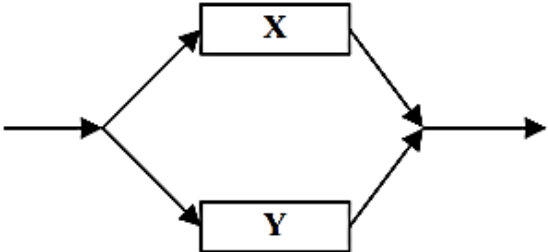
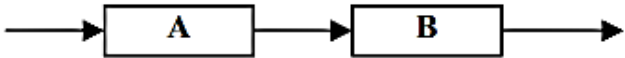
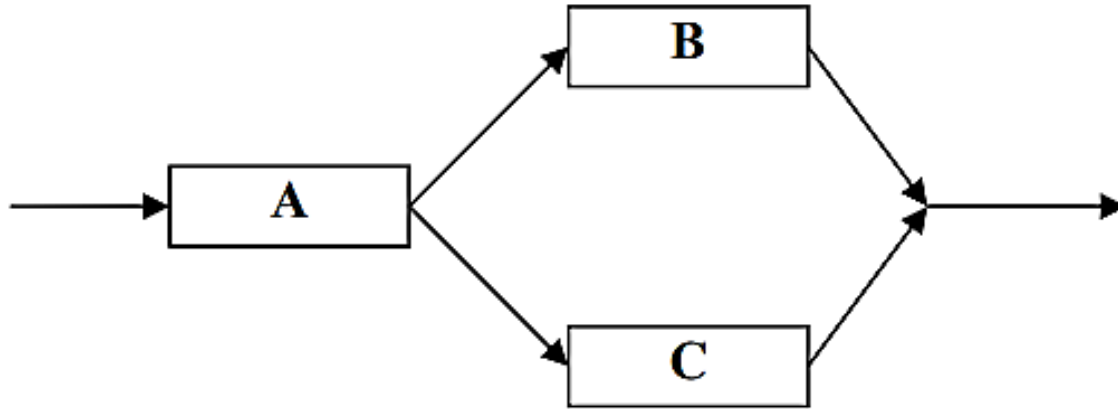
You see something matching this ...	Then solve using this ...
	GAR: $P(X \text{ OR } Y) = P(X) + P(Y) - P(X)*P(Y)$
	BMR: $P(A \text{ AND } B) = P(A) * P(B)$

Table 1: The basic subsystems for all solutions.

The probability that components (or people or circuits or whatever) A, B working is 0.8, 0.9 respectively. The probability that component C failing is 0.4, and the probability of component D failing is 0.15. Each A,B,C,D is independent



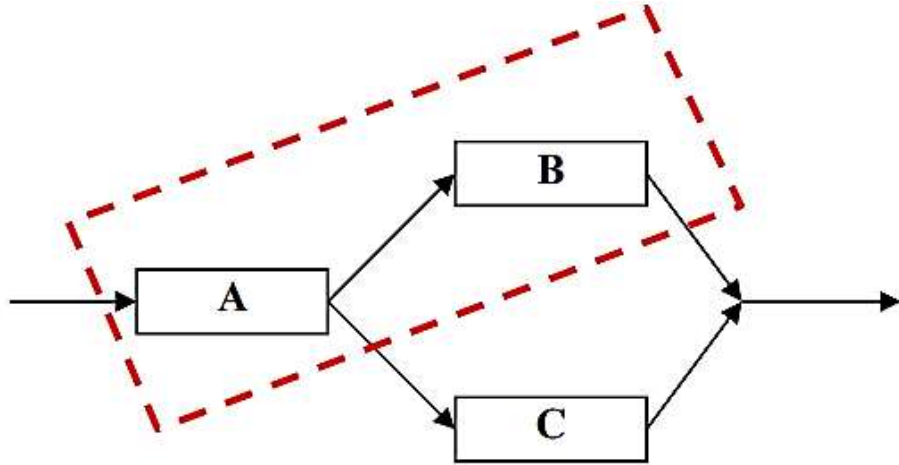


Figure 1: Matching like this is not allowed!

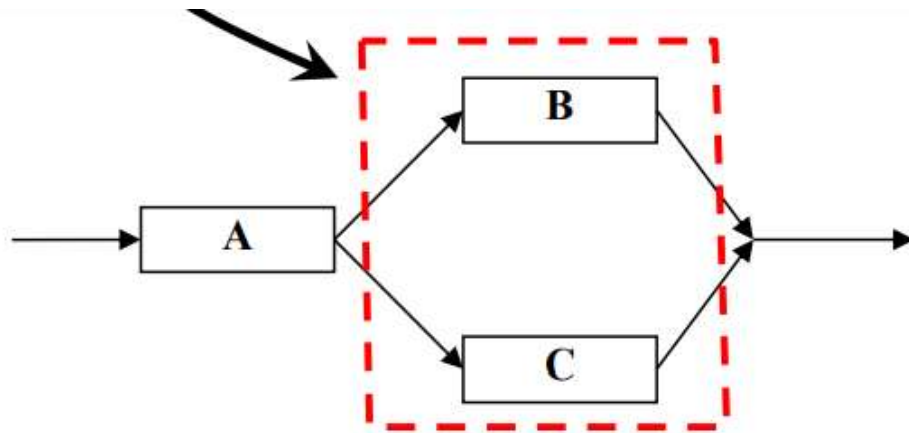


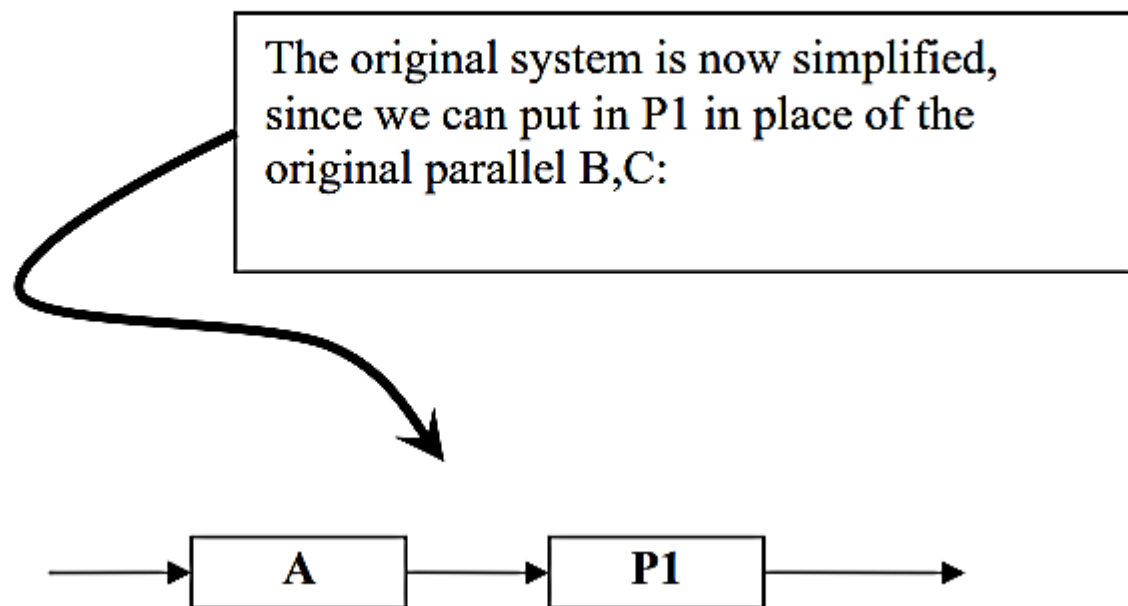
Figure 2: Matching like this is what we are looking for.

So working only now on this simple P1 system,
 $P(\text{P1 works}) = P(\text{(EITHER) B works OR C works})$

$$\begin{aligned} &= P(\text{B works}) + P(\text{C works}) - P(\text{B works AND C works}) && [\text{general addition rule}] \\ &= 0.9 + 0.6 - P(\text{B works AND C works}) && [P(\text{C works}) = 1 - P(\text{C fails}) = 0.6] \end{aligned}$$

$$\begin{aligned} &= 1.5 - P(\text{B works}) * P(\text{C works}) && [\text{basic multiplication rule}] \\ &= 1.5 - 0.9 * 0.6 \\ &= 0.96 \end{aligned}$$

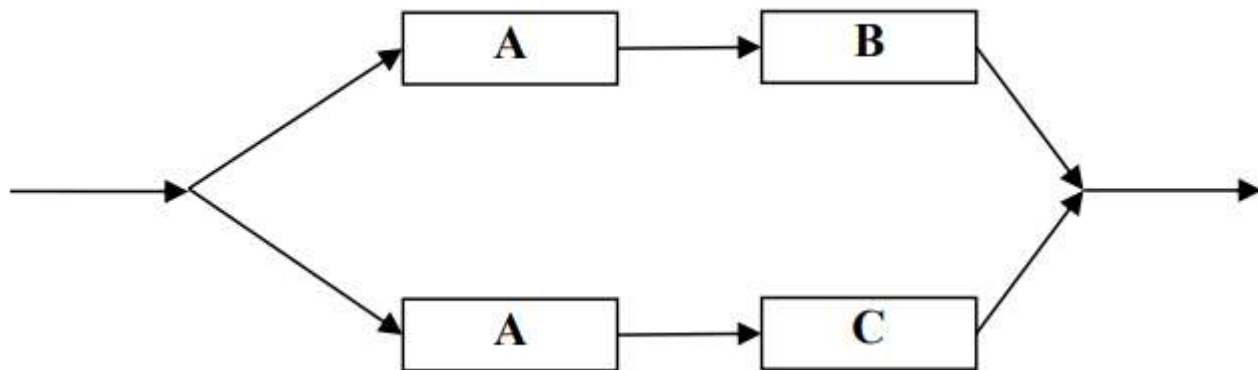
With this we can now redraw the original system a bit simpler:





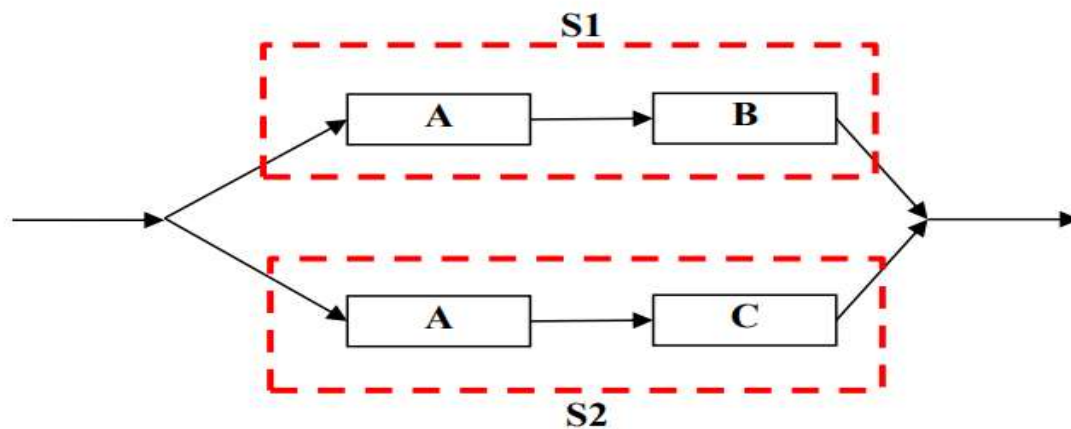
Now this is just like a “system 2”, so can be solved using the basic multiplication rule:

$$\begin{aligned} P(\text{System works}) &= P(\text{(BOTH) A works AND P1 works}) \\ &= P(\text{A works}) * P(\text{P1 works}) && \text{[basic multiplication rule]} \\ &= 0.8 * 0.96 \\ &= 0.768 \end{aligned}$$



Answer Details:

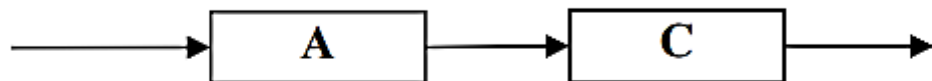
Break it down ... this is really the same as:



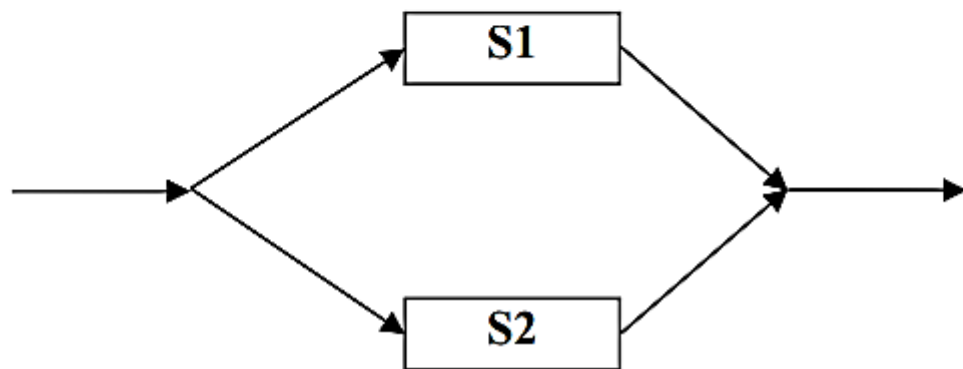
Where S1 is:



And S2 is:



And the overall system is now:



$$P(\text{S1 works}) = 0.72$$

$$P(\text{S2 works}) = 0.48$$

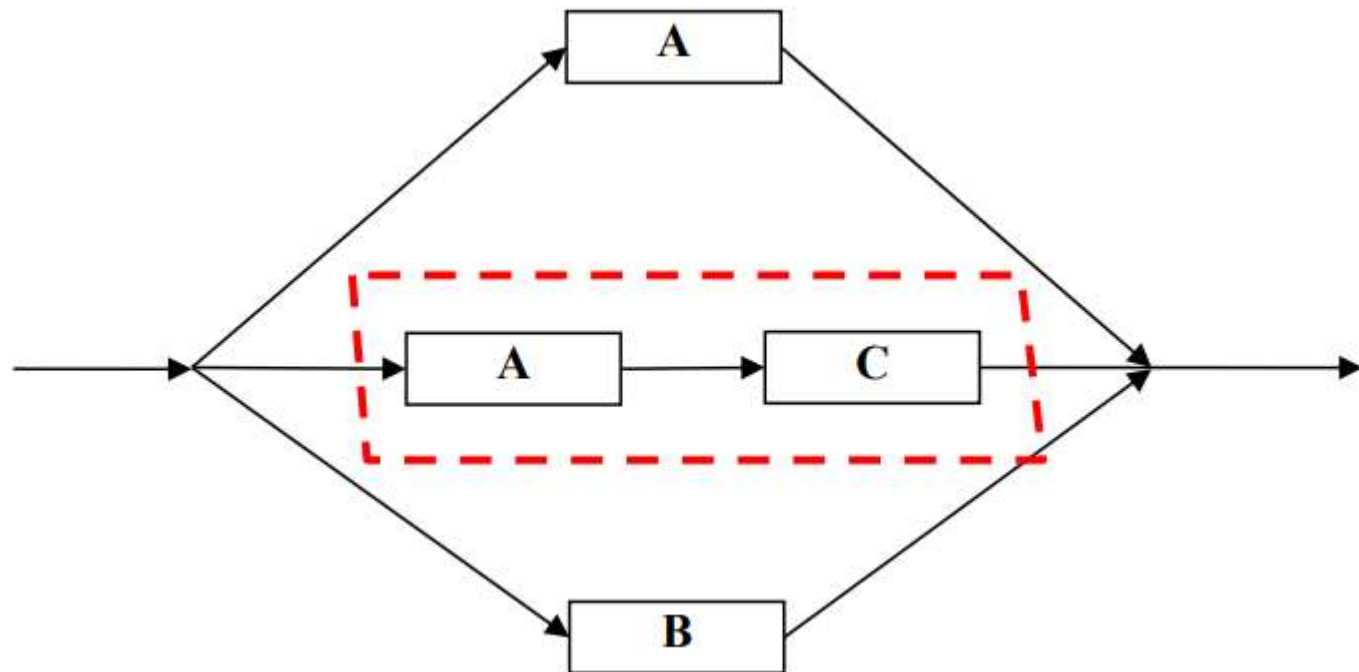
$$P(\text{System works}) = P(\text{S1 works } \underline{\text{OR}} \text{ S2 works})$$

$$= P(\text{S1 works}) + P(\text{S2 works}) - P(\text{S1 works } \underline{\text{AND}} \text{ S2 works})$$

$$= P(\text{S1 works}) + P(\text{S2 works}) - P(\text{S1 works}) * P(\text{S2 works}) \quad [\textit{using basic mult rule}]$$

$$= 0.72 + 0.48 - 0.72 * 0.48$$

$$= 0.8544.$$



$$P(S1 \text{ works}) = 0.72$$

$$P(S2 \text{ works}) = 0.48$$

Next simplify the original system with the new information to:

$$P(S1 \text{ works}) = P(A \text{ works } \underline{\text{AND}} C \text{ works}).$$

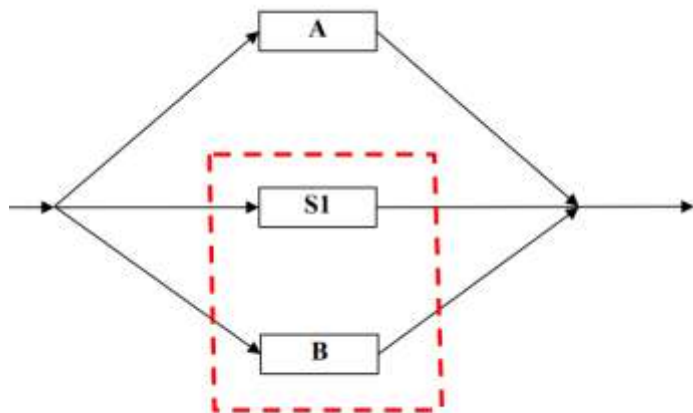
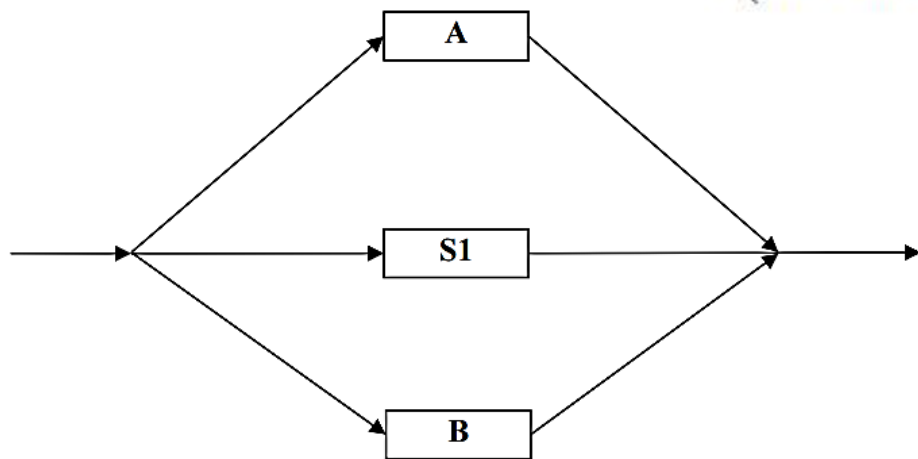


Figure 4: Another possible "system 1" type sub-system.

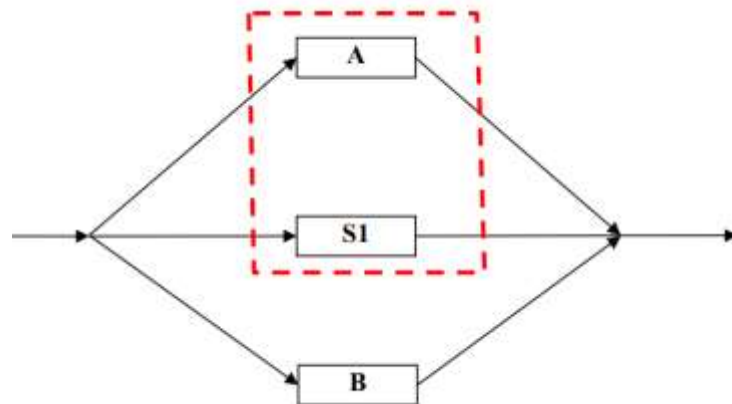


Figure 3: Picking out one possible "system 1" type sub-system.

Say I take the second option here, and call the new sub-system P1, then:

$$P(\text{P1 works}) = P(\text{(EITHER) S1 works OR B works})$$

$$= P(\text{S1 works}) + P(\text{B works}) - P(\text{S1 works AND B works}) \quad [\text{general addition rule}]$$

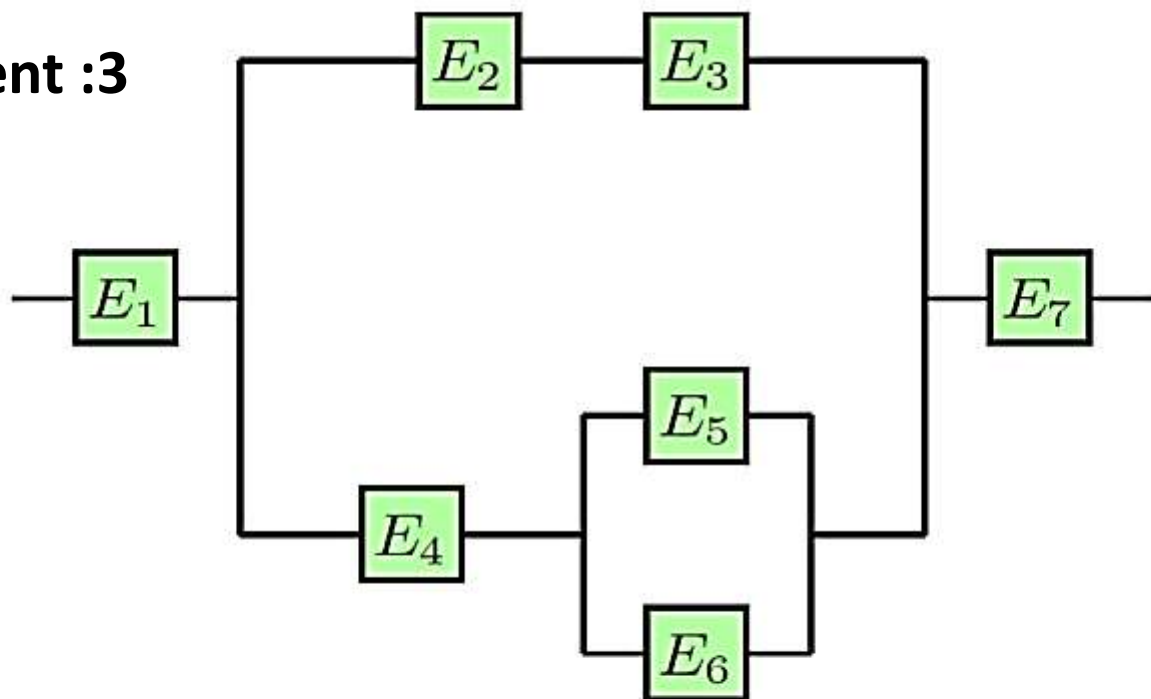
$$= 0.48 + 0.9 - 0.48 * 0.9 \quad [\text{applying basic multiplication rule for second bit}$$

here]

$$= 0.948$$

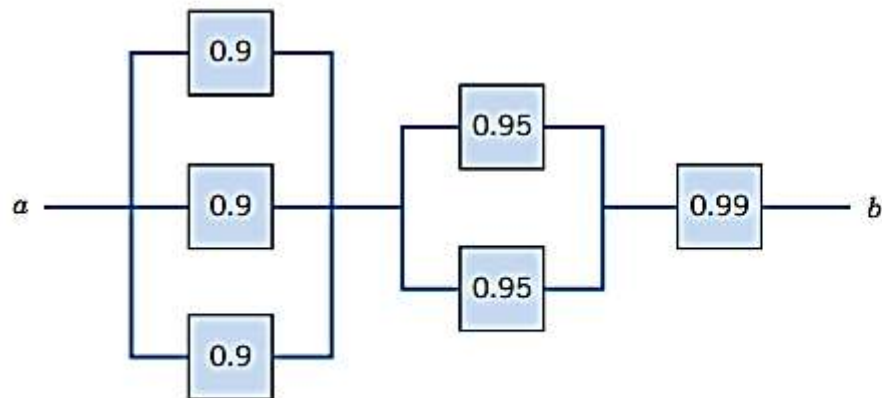
The system now simplifies to:

Assignment :3

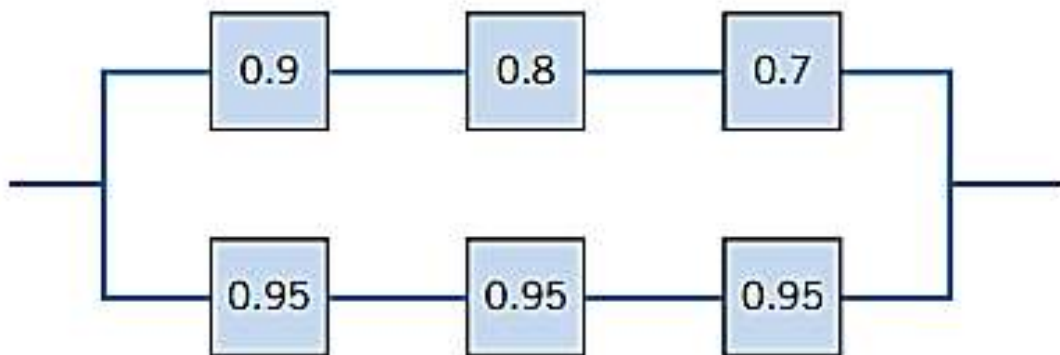


Component	E_1	E_2	E_3	E_4	E_5	E_6	E_7
Probability of functioning well	0.9	0.5	0.3	0.1	0.4	0.5	0.8

The following circuit operates only if there is a path of functional devices from left to right. The probability that each device functions is shown on the graph. Assume that devices fail independently. What is the probability that the circuit operates?



2-90. The following circuit operates if and only if there is a path of functional devices from left to right. The probability that each device functions is as shown. Assume that the probability that a device is functional does not depend on whether or not other devices are functional. What is the probability that the circuit operates?



The following circuit operates if and only if there is a path of functional devices from left to right. The probability each device functions is as shown. Assume that the probability that a device function,

